INCH-POUND
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15 April 1994

PERFORMANCE SPECIFICATION

AIR CLEANER, ENGINE

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, test, manufacture and acceptance requirements for the engine air cleaner, referred to herein as the air cleaner, and the replaceable air filter element, referred to herein as the filter element. Unless indicated otherwise, references to the air cleaner can be understood to include the filter element.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other section of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2940

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2.2 Government documents.

2.2.1 Government documents, drawings, and publications. The following Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

19207-12307204	- Air Cleaner, Engine (Interface)
19207-12329100	- Element, Air Filter (Interface)

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NCSL Z540.1	- General Requirements for Calibration Laboratories and Measuring and Test Equipment.

(Application for copies should be addressed to American National Standards Institute, 11 West 42nd Street, New York 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B633	- Electrodeposited Coatings of Zinc on Iron and Steel (DoD Adopted).

(Application for copies of ASTM publications may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J726	- Air Cleaner Test Code, Recommended Practice (DoD Adopted).

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(Application of copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Design, materials, and manufacturing processes. Unless otherwise specified, the design, materials, and manufacturing process selection shall be the prerogative of the contractor as long as all articles submitted to the Government fully meet the operating, interface, ownership and support, and operating environment requirements specified. (Asbestos and cadmium materials shall not be used in any form in any part of the air cleaner or filter element. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcuries per gram or activity per item equals or exceeds 0.01 microcuries).

3.2.1 Dissimilar Metals. The air cleaner and filter element shall be designed and fabricated to minimize galvanic corrosion in dissimilar metal couples.

3.2.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Operating requirements. The air cleaner shall meet the following performance requirements:

3.3.1 Air leakage. The air cleaner, when subjected to a vacuum of 50 inches of water or a pressure of 2 pounds per square inch (psi) (13.8 kilopascals (kPa)) for a period of not less than 5 minutes (min), shall not leak more than 0.3 standard cubic feet per minute ($stdft^3$/min) (0.141 liter per second (L/s)).

NOTE: This requirement does not apply to the filter element.

3.3.2 Airflow restriction. Pressure drop from intake to outlet shall not exceed 9.6 in. of water (2.38 kPa) when filter is clean and airflow is 980 ft^3/min. (462.5 L/s).

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3.3.3 Initial efficiency. Air cleaner initial efficiency shall be 99.5 percent (%) minimum, when installed.

3.3.4 Dust capacity. Dust capacity shall be such that the pressure drop across the air cleaner shall not exceed 20 in. of water (4.97 kPa) after 28 hours (hr) of operation with a vacuum of 5 in. of water (1.24 kPa) or a continuous air flow of 98 ft^3/min. (46.2 L/s) at the scavenging outlet and dust continuously introduced at a rate of 0.025 gram per cubic foot (g/ft^3) (0.88 gram per cubic meter (g/m^3)) of air. Particle size distribution shall be as specified in SAE J726.

3.3.5 Airflow restriction after cleaning. Pressure drop across the air cleaner shall not exceed 1 in. of water (0.25 kPa) increase over original pressure drop after the air cleaner has been operated to dust capacity, filter element cleaned and the same filter element reinstalled.

3.3.6 Flame resistance. Filter element shall not continue to burn or smolder for more than 10 seconds after removal of an applied flame.

3.4 Interface requirements.

3.4.1 Overall envelope & interfaces. Envelope dimensions, and the physical interface dimensions and mounting locations of the air cleaner and the filter element, shall conform to drawings 19207-12307204 and 19207-12329100, respectively.

3.4.2 Air cleaner.

3.4.2.1 Weight. Weight of the air cleaner including a new filter element shall be not greater than 100 pounds (lb) (45.4 kilograms (kg)).

3.4.2.2 Tubular joints. The tubular ends of the air cleaner inlet and outlet and the scavenging outlet shall have externally beaded ends as specified in figure 1.

3.4.3 Filter element. The filter element shall be of a one-piece design, firmly bolted or clamped in place and easily replaced using hand clamps or common hand tools. The filter element is reusable and capable of multitude cleanings.

3.4.3.1 Weight. The weight of the filter element shall not exceed 30 lb (13.6 kg).

3.5 Ownership and support requirements. The air cleaner shall meet the following logistics requirements.

3.5.1 Identification and marking. Unless otherwise specified (see 6.2), the air cleaner shall be permanently and legibly marked on a nameplate, and shall include, as a minimum, the following:

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a. Nomenclature “AIR CLEANER, ENGINE”.
b. Manufacturer’s identification (CAGE).
c. Military part number.
d. Manufacturer’s serial number or part number.
e. Date of manufacture.
f. Symbol “US”.

3.6 Operating environment requirements. The air cleaner shall operate under the following environmental conditions, without degradation in performance:

3.6.1 Shock. The air cleaner shall demonstrate no performance degradation and show no evidence of damage or deformation after being subjected to shock pulses of 40 gravity units (g) for 11 millisecond (ms) time duration applied along each of the three major axes.

3.6.2 Vibration. The air cleaner shall demonstrate no performance degradation and show no evidence of damage or deformation after being subjected to vibration levels, in each of the three major axes, of 1.5 and 4.2 g in the frequency range of 5 to 500 Hertz (Hz) with a sweep time of 15 min.

3.6.3 Steam and waterjet cleaning. The air cleaner shall demonstrate no performance degradation and show no evidence of damage or deformation when subjected to steam cleaning, using high pressure steam and commercial heavy duty cleaner followed by waterjet cleaning.

3.6.4 Protective coating. A protective finish shall be applied to the exterior and interior surfaces of the air cleaner. The filter element carrier and latch shall be zinc plated in accordance with ASTM B633, type II, SC-2 or equivalent. The zinc plating shall be a minimum of 0.5 mils thick. Stainless steel springs of the retainer assembly, when used shall not require protective coating.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspection (see 4.1.2).

4.1.1 First article inspection. When specified (see 6.2 and 6.3), first article inspection shall be performed on three air cleaners and filters. First article inspection shall include all verifications of table I.

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TABLE I. Verification methods.

Title	Requirement	Verification
Materials & dissimilar metals	3.2 & 3.2.1	4.2
Operating requirements	3.3	4.2.3
Air leakage	3.3.1	4.2.3.1
Airflow restriction	3.3.2	4.2.3.2
Initial efficiency	3.3.3	4.2.3.3
Dust capacity	3.3.4	4.2.3.4
Airflow restriction after cleaning	3.3.5	4.2.3.5
Flame resistance	3.3.6	4.2.3.6
Interface requirements	3.4	4.2.4
Overall envelope	3.4.1	4.2.4.1
Air cleaner	3.4.2	4.2.4.2
Filter element	3.4.3	4.2.4.3
Ownership & support requirements	3.5	4.2.5
Name plate & product marking	3.5.1	4.2.5.1
Operating environment requirements	3.6	4.2.6
Shock	3.6.1	4.2.6.1
Vibration	3.6.2	4.2.6.2
Steam & waterjet cleaning	3.6.3	4.2.6.3
Protective coating	3.6.4	4.2.6.4

4.1.2 Conformance inspection. Conformance inspection shall consist of those examinations and tests from table I as defined in the contract.

4.2 Verification methods. Acceptable verification methods included in this section are visual inspection, and measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

4.2.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures to verify performance. See the contract for alternatives that replace verification methods required by this specification.

4.2.2 Test conditions and equipment.

4.2.2.1 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 77 ±18°F (25 ± 10°C)
b. Atmospheric pressure: Site pressure
c. Relative humidity: Uncontrolled

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4.2.2.2 Test conditions.

4.2.2.2.1 General. Test specimens shall be stabilized at room temperature for 1 hr before tests are begun.

4.2.2.2.2 Temperature and humidity. All performance tests shall be conducted with air entering the air cleaner at a temperature of 75 ± 15°F (24 ± 8°C) and a relative humidity of 30 to 65%.

4.2.2.2.3 Dust feeding rate. When the feeding of dust is required as part of any test, the dust shall be fed at an average rate of 0.025 g/ft^3 (0.88 g/m^3) of air. The dust rate shall not vary more than ± 0.005 g/ft^3 (0.18 g/m^3) during any part of the applicable test.

4.2.2.2.4 Test dust. Test dust shall be of two grades labeled fine and coarse. A typical chemical analysis is given in accordance with SAE J726. Particle size distribution shall be as specified in SAE J726.

4.2.2.2.5 Airflow. Specified values of airflow are for standard atmosphere conditions of 29.92 in. of mercury (101.3 kPa) barometric pressure and 80°F (27°C) ambient temperature providing an air density of 0.073 pound per cubic foot (lb/ft^3) (1.17 kilogram per cubic meter (kg/m^3)). Actual airflow values during testing shall be corrected to standard conditions and shall be within ± 2% of specified values.

4.2.2.2.6 Airflow restriction. Actual airflow restriction values shall be corrected to airflow with a density of 0.073 lb/ft^3 (1.17 kg/m^3) and shall be accurate to ± 0.1 in. of water (± 0.0248 kPa).

4.2.2.2.7 Air cleaner mounting position. The air cleaner shall be mounted in its normal operating attitude and shall be electrically grounded (see figure 2).

4.2.2.3 Test equipment.

4.2.2.3.1 Piezometer tubes, transition ducts, adapter, manometer and flowmeter. Piezometer tubes conforming to figure 3 shall be installed at the inlet and outlet of the air cleaner. The inside diameter of the piezometer tubes shall be equal to the inside diameter of the air cleaner outlet (+0.5, -0.0 in.) (+1.27,-0.0 centimeters (cm)). Transition ducts shall be installed between the air cleaner inlet and inlet piezometer tube. Transition ducts and adapters shall have sides at angles no greater than seven degrees measured from the centerline of the duct or adapter. A smooth approach (figure 4) shall be installed at the front of the inlet piezometer tube. A water manometer shall be installed between the piezometer tubes (see figure 2). A flowmeter shall be connected to the outlet of the absolute filter.

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4.2.2.3.1.1 Accuracy of test instrumentation calibration. The accuracy of instruments and test equipment used to control or monitor the test parameters shall be verified prior to and following each test and then calibrated in predetermined intervals and shall meet the requirements of ANSI Z540.1 and SAE J726 as applicable. All instruments and test equipment used in conducting the tests specified herein shall:

a. Be calibrated to laboratory standards whose calibration is traceable to the National Standards via primary standards.
b. Have an accuracy of at least one-third the tolerance for the variable to be measured. In the event of conflict between this accuracy and a requirement for accuracy in any one of the test methods of this standard, the latter shall govern.

4.2.2.3.2 Absolute filter. The absolute filter shall consist of one layer of 0.50 in. (1.27 cm) fiberglass batting conforming to SAE J726 installed in a suitable housing with the nap side of the fiberglass batting facing upstream. The filter batting shall be of sufficient size to prevent excessive restriction. The face velocity of air against the filter batting shall not exceed 165 feet per minute (ft/min.) (838 millimeters per second (mm/s). A manometer shall be located in the test setup to measure the pressure drop across the filter (see figure 2). A new filter batting shall be installed if, during the dust capacity test (see 3.3.4 and 4.2.3.4), the air pressure drop across the filter increases by 10 in. of water above its original restriction. The filter housing shall be electrically grounded during tests.

4.2.2.3.3 Drying and oven-weighing of absolute filter. The fiberglass batting shall be hung in an air oven and dried for 6 hr at temperatures of 200 to 240°F (93.3 to 115°C). Drying time may be reduced, and the batting considered completely dry if there is no change in the weight of the batting in two consecutive 10 min intervals. All weighings shall be done with the fiberglass batting in the oven**.** The drying and oven-weighing procedure shall be observed before and after each test involving dust feeding.

4.2.2.3.4 Scavenge vacuum. During all tests, except dust capacity test, a scavenge vacuum of 5 in. of water (1.24 kPa) or a continuous air flow of 98 ft^3/min. (46.2 L/s) shall be applied to the scavenge outlet. A piezometer tube of suitable capacity, a manometer and a dust collecting system shall be connected to the scavenge outlet (see figure 2).

4.2.2.3.5 Dust feeder. The dust feeder (see figure 5) shall be of a type that will deliver dust to the cleaner without breaking down the dust particles or creating conglomerates. The feeder shall deliver the dust uniformly throughout the test. Dust at the beginning of testing shall have the same particle distribution as dust at the conclusion of testing.

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4.2.3 Operating requirements verification.

4.2.3.1 Air leakage. To determine conformance to 3.3.1, the air cleaner inlet shall be connected to a flowmeter and all other inlets and outlets shall be sealed to prevent air leakage. The flowmeter shall be as accurate as a 0.19 in. (4.8 mm) orifice connected to a ±0.01 in. (0.25 mm) water manometer. The flowmeter shall indicate an airflow no greater than 0.3 ft^3/min. (.0141 L/s) when the air cleaner is subjected to a vacuum of 50 in. of water (12.44 kPa) when measured at the air cleaner air outlet. As an alternate method, the air cleaner shall be pressurized at not less than 2 psi (13.8 kPa) pressure. Air leakage shall not exceed the specified limit.

4.2.3.2 Airflow restriction. To determine conformance to 3.3.2, using the air cleaner set-up as shown in figure 2, airflow restriction in in. of water shall be determined at the rated airflow of 980 ft^3/min. (462.5 L/s).

4.2.3.3 Initial efficiency. To determine conformance to 3.3.3, the efficiency shall be determined by the method specified in SAE J726 after 60 min of testing. The following test shall be conducted with the air cleaner set up as shown in figure 2, the airflow shall be adjusted to 980 ft^3/min. (462.5 L/s). Fine grade dust in accordance with 4.2.2.2.4 shall be fed in accordance with 4.2.2.2.3 for a period of 60 min with airflow restriction recorded at start of test, and at 10 min intervals. The absolute filter shall be dried and oven-weighed in accordance with 4.2.2.2.3 taking care to retain all entrapped dust. Weights shall be accurate to 0.01 gram, wherever possible, but never less accurate than 0.1 gram. Initial efficiency shall be determined by means of the following equation:

$$\text{Efficiency in percent} = \frac{W_1 - (W_2 - W_3) \times 100}{W_1}$$

Where: W_1 = is weight of dust fed into air cleaner inlet.
W_2 = is the weight of absolute filter and entrapped dust.
W_3 = is weight of absolute filter before tests.

Air cleaner efficiency below 99.5 percent, shall be cause for rejection of items.

4.2.3.4 Dust capacity. To determine conformance to 3.3.4, the following dust capacity test shall be conducted. A new filter element shall be installed for this test. With the air cleaner set up as shown in figure 1, the airflow shall be adjusted to 980 ft^3/min. (462.5 L/s). Airflow restriction shall be recorded at the start of testing and at 10 min intervals throughout the test. Coarse grade dust in accordance with 4.2.2.2.4 shall be fed in accordance with 4.2.2.2.3. After obtaining the initial airflow restriction at rated airflow, the airflow shall be varied as shown in figure 5. Figure 5 is based on SAE J726 and will result in an average airflow which is 60% of rated airflow. This airflow cycle shall be continued for the duration of this test. The dust capacity test shall be terminated when airflow restriction exceeds 20 in. of water (4.97 kPa), or at the end of 28 hours, whichever occurs first, and shall be cause for rejection.

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4.2.3.5 Airflow restriction after cleaning. To determine conformance to 3.3.5, after the dust capacity test of 4.2.3.4, the filter element shall be removed from the air cleaner and cleaned with compressed air as follows. The compressed air shall be applied first in a direction opposite to normal airflow, and then in the direction of normal airflow. Line air pressure shall be 100 ±10 pounds per square inch gage (psig) (689 ±69 kPa), and the nozzle inside diameter shall be nominal 0.12 in. (0.30 cm). After the filter element is thoroughly cleaned, it shall be reinstalled in the air cleaner. The airflow shall be adjusted to 980 ft^3/min. (462.5 L/s) and the airflow restriction shall be determined. A pressure drop across the air cleaner that exceeds 1 in. of water (0.25 kPa) shall be cause for rejection of items.

4.2.3.6 Flame resistance. To determine conformance to 3.3.6, a sample of the filter medium used in the filter element shall be exposed to a flame until the sample burns or glows. The flame shall be removed and the filter element shall be observed. A continuous burn or smolder of the filter element for more than 10 seconds shall be cause for rejection of items.

4.2.4 Interface requirements verification.

4.2.4.1 Overall envelope inspection. Demonstrate placement of the air cleaner and the filter element in the present existing location on a military tracked vehicles in the 500 HP A0, A1 and A2 Bradley Fighting Vehicle (BFV) family.

4.2.4.2 Air cleaner.

4.2.4.2.1 Tubular joints (air cleaner). Inspect the tubular joints for compliance as specified.

4.2.4.2.2 Weight (air cleaner). Weigh the air cleaner to be 100 lb (45.5 kg) or less.

4.2.4.3 Filter element. To determine conformance to 3.4.3, inspect the filter element for conformance as specified herein.

4.2.4.3.1 Weight (filter element). Weigh the filter element to be 30 lb (13.6 kg) or less.

4.2.5 Ownership and support requirements verification.

4.2.5.1 Identification inspection. Verify the presence of the required marking on the air cleaner. After performing all environmental test in 4.2.6, re-inspect the air cleaner markings for readability.

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4.2.6 Operating environment requirements verification.

4.2.6.1 Shock. To determine conformance to 3.6.1, the air cleaner shall be mounted in a manner that simulates actual use, or shall be rigidly mounted to a shock table, and shall be subjected to three sawtooth shocks of 40 g in accordance with figure 7 and time duration of 11 ms applied along the three mutually perpendicular axes of the air cleaner for a total of 18 shocks. No performance testing shall be required during shock. At the conclusion of shock testing, the air cleaner shall be subjected to the air leakage and initial efficiency tests specified in 4.2.3.1 and 4.2.3.3.

4.2.6.2 Vibration. To determine conformance to 3.6.2, the air cleaner shall be mounted in a manner that will simulate its actual use and shall be vibrated along each axis in accordance with the vibration levels specified in figure 8. The sweep time shall be 15 min for the sweep frequency range of 5 to 500 to 5 Hz. The cycling time shall not exceed 3 hr in each axis. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 9. The specified sweep time is that of an ascending plus a descending sweep and is twice the ascending sweep time shown in figure 9 for the specified range. When the air cleaner resonances below 5 Hz are measured , the test curve shall be extended to 2 Hz and the sweep time shall be 18 min (2 to 500 to 2 Hz). There shall be no operational tests required during vibration. After the test, the air cleaner shall be subjected to the air leakage and initial efficiency tests specified in 4.2.3.1 and 4.2.3.3.

4.2.6.3 Steam and waterjet cleaning. To determine conformance to 3.6.3, with the inlet and outlet tubes suitably sealed, the air cleaner shall be subjected to steam cleaning, followed immediately by waterjet cleaning. The jet shall be applied perpendicular to the surface being cleaned at a distance of not greater than 1 foot (ft) (0.31 meters (m)) for steam cleaning, and not greater than 3 ft (0.91 m) for waterjet cleaning. The jet pressure shall be 105 $\pm$ 5 psig (724 $\pm$ 34 kPa) for steam cleaning and 50 $\pm$ 5 psig (345 $\pm$ 34 kPa) for waterjet cleaning. The jet shall be moved at such a rate that the exposed surface of the air cleaner is subjected to the jet at the rate of 1 square foot per minute (ft^2/min) (0.092 square meter per minute (m^2/min) for 2 to 3 min. Prior to and within 2 hr after completing the steam and waterjet cleaning, the air cleaner shall be subjected to the air leakage and initial efficiency tests specified in 4.2.3.1 and 4.2.3.3. Moisture inside the air cleaner shall not be considered a failure.

4.2.6.4 Protective coating inspection. Inspect the protective coating for conformance as specified herein.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite

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packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Air cleaners manufactured in accordance with this specification are intended for use on military tracked vehicles in the 500 HP A0, A1 and A2 Bradley Fighting Vehicle (BFV) Family. Air cleaner is military unique because it must withstand the dust conditions and the types of high shock and vibration that would be experienced on the battlefield, that exceed any commercial requirements.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. If the first article is required (see 3.1).
d. Identification and marking if other than as specified (see 3.5.1).
e. If inspection conditions should be other than as specified (see 4.2.2.1).
f. If sample size for first article inspection is other than as specified (see 4.1.1).
g. Packaging requirements (see 5.1).

6.3 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, Government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those

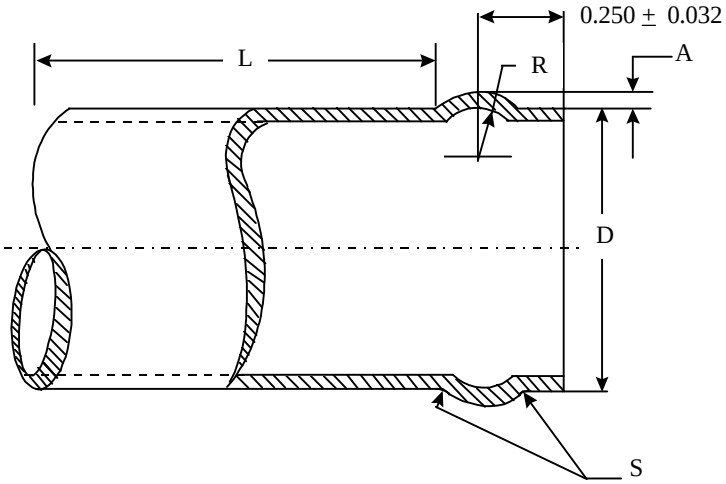
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processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table I and their designated frequency based on a risk assessment for the procurement.

6.5 Subjected term (key word) listing.

Absolute filter	Leakage
Airflow	Military tracked vehicles
Bradley fighting vehicle family	Piezometer
Dust	Scavenge vacuum
Flame resistant	

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



Tubular End / Air Cleaner	D Tube OD	Tolerance for Length L	A + 0.003 -0.010 Bead Height	L Min. Length	R Max. Rad	S Max. Rad
Scavenging Outlet	1.500	+0.010 -0.020	0.062	0.750	0.156	0.093
Outlet	6.000	± 0.020	0.077	0.750	0.156	0.093
Inlet	9.000	± 0.020	0.077	0.750	0.156	0.093

NOTE: All dimensions are in inches

FIGURE 1. Tubular joints.

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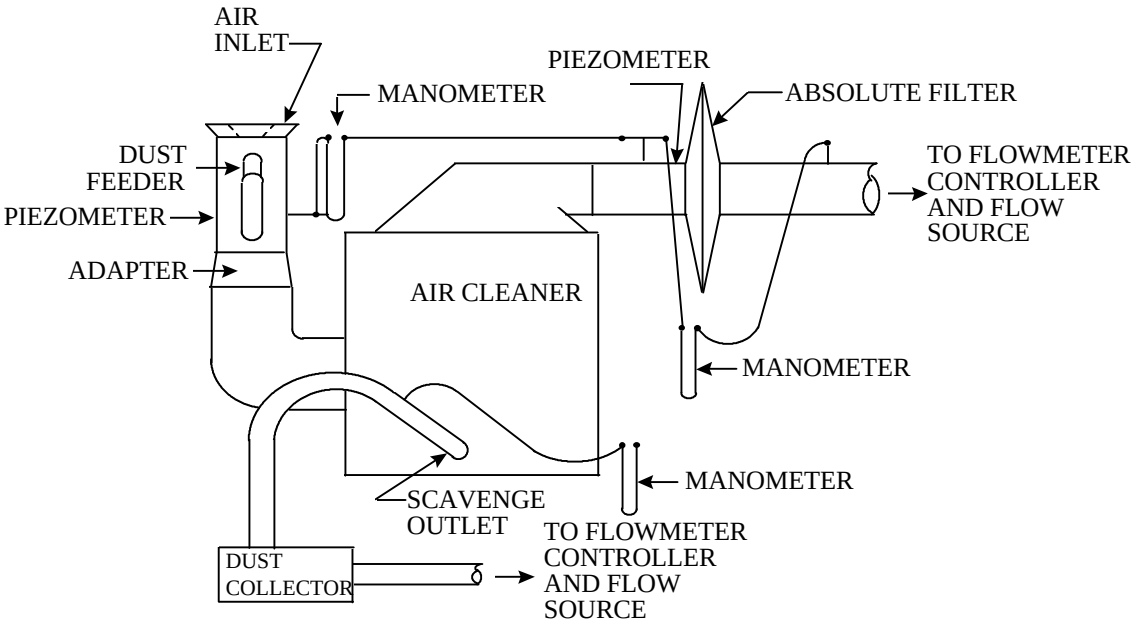
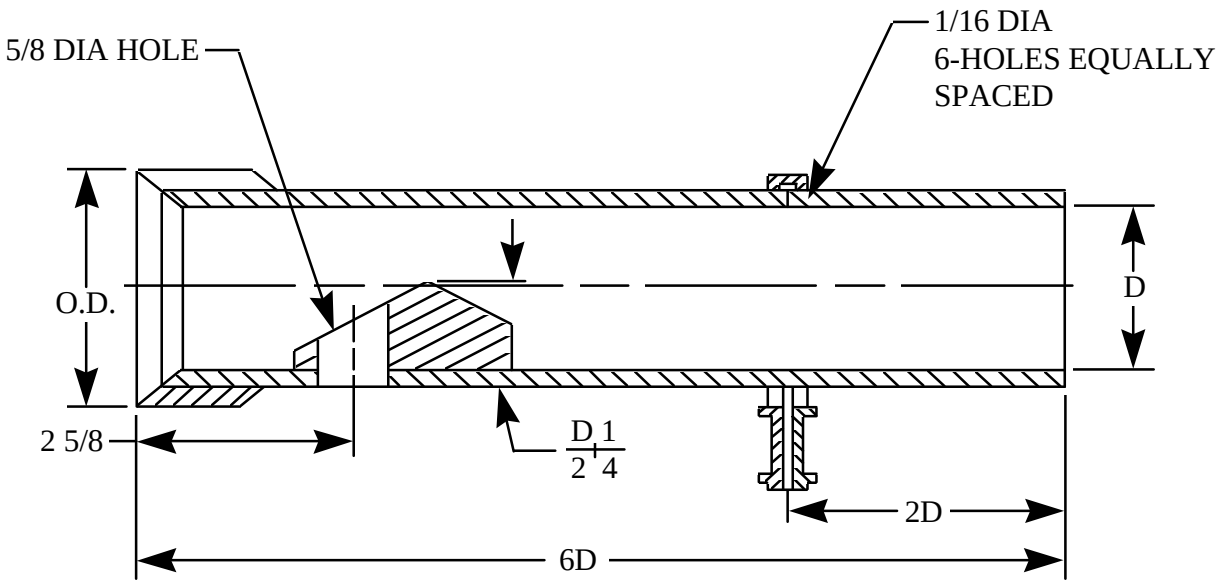
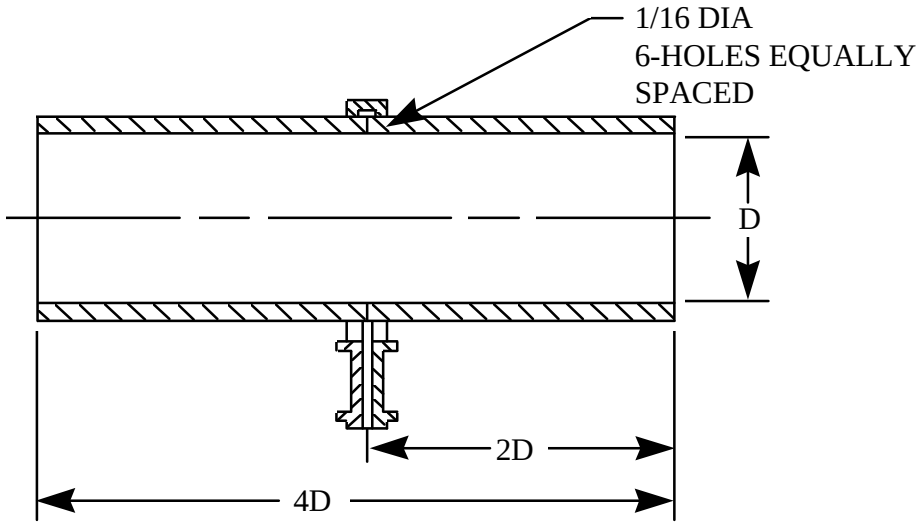


FIGURE 2. Test setup.

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INLET PIEZOMETER TUBE



OUTLET PIEZOMETER TUBE

FIGURE 3. Piezometer tubes.

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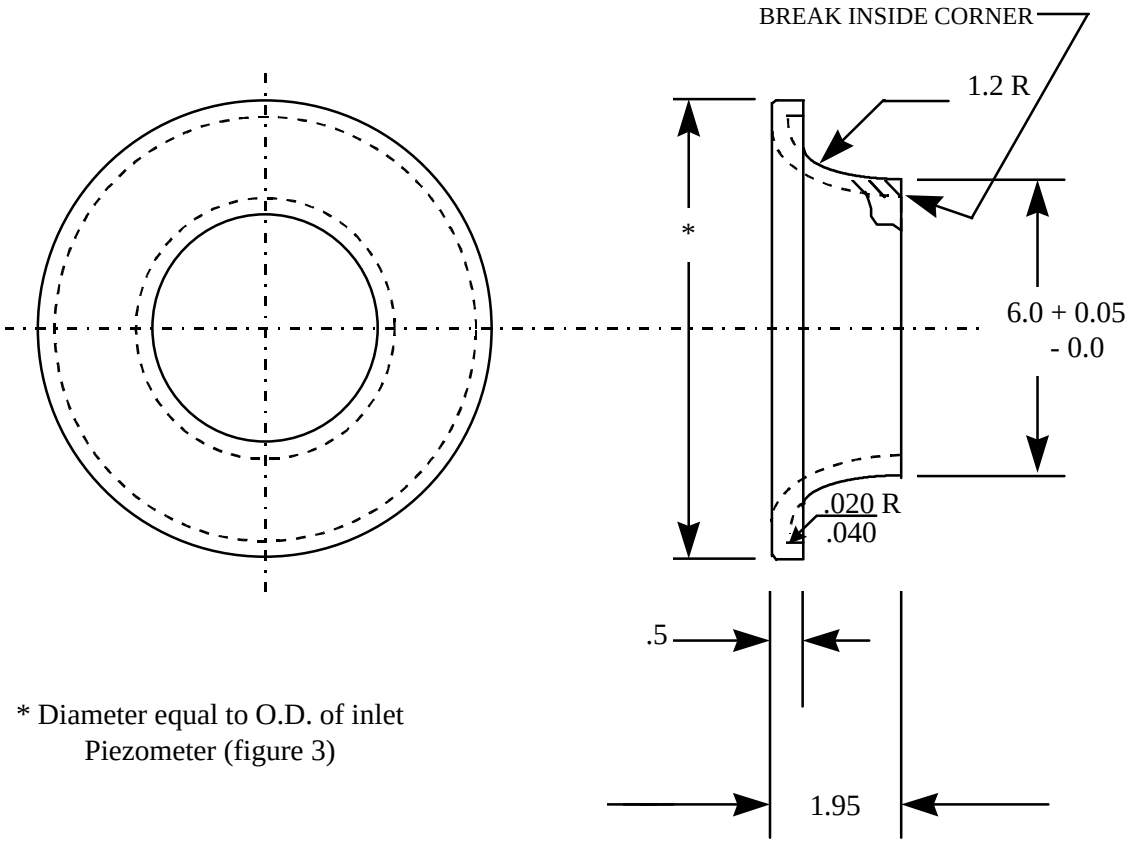


FIGURE 4. Ideal flow nozzle.

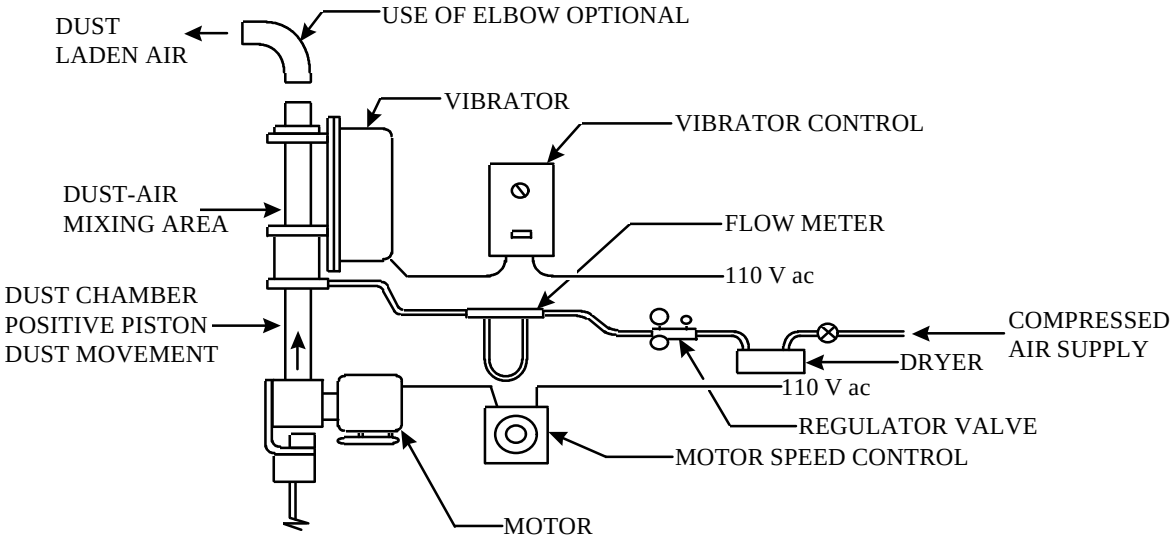


FIGURE 5. Dust feeder.

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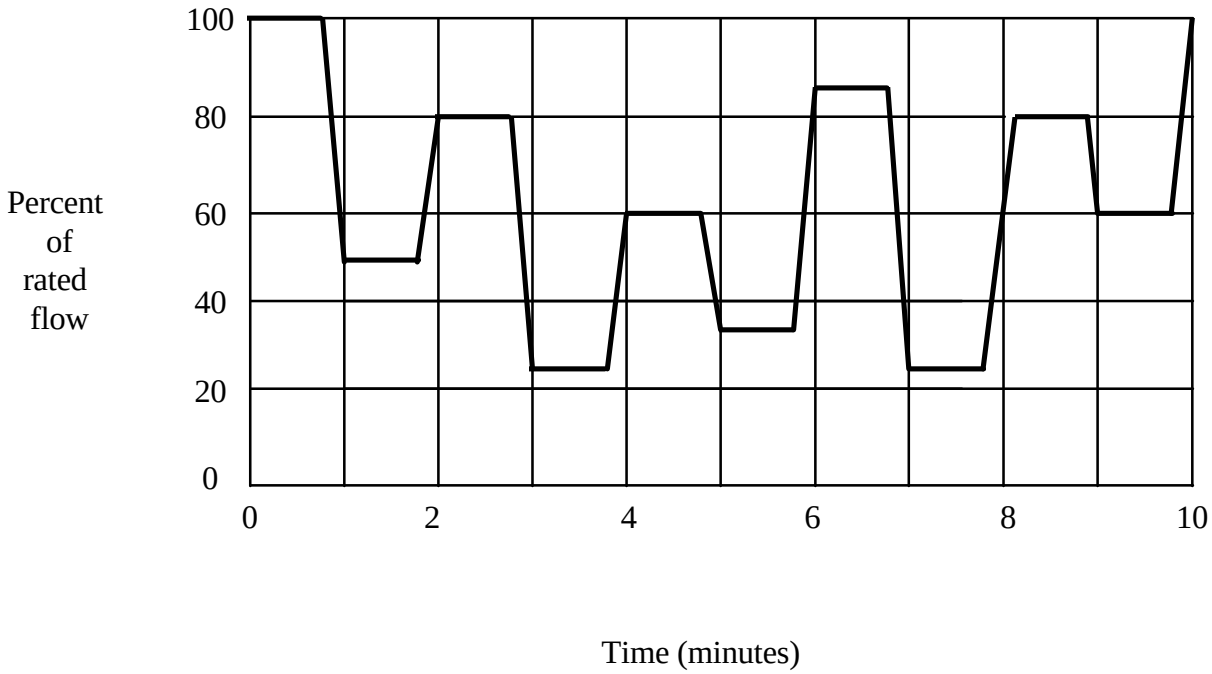
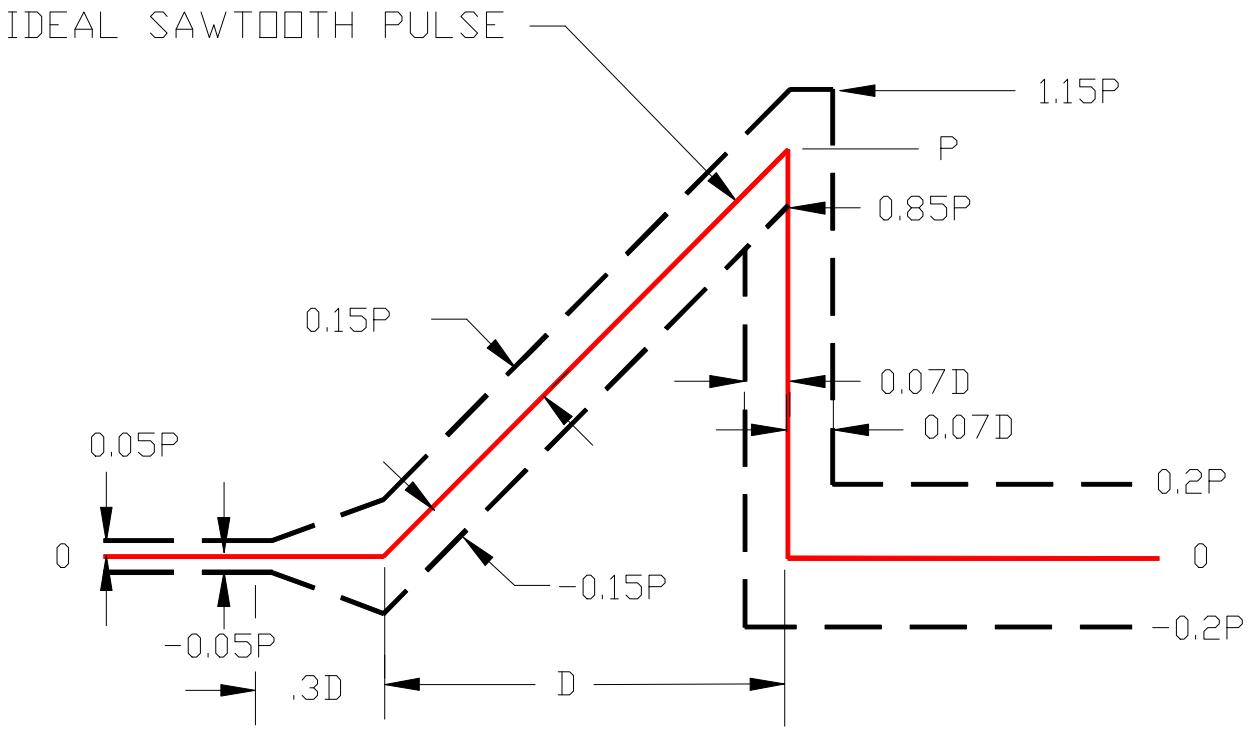


FIGURE 6. Airflow vibrations.

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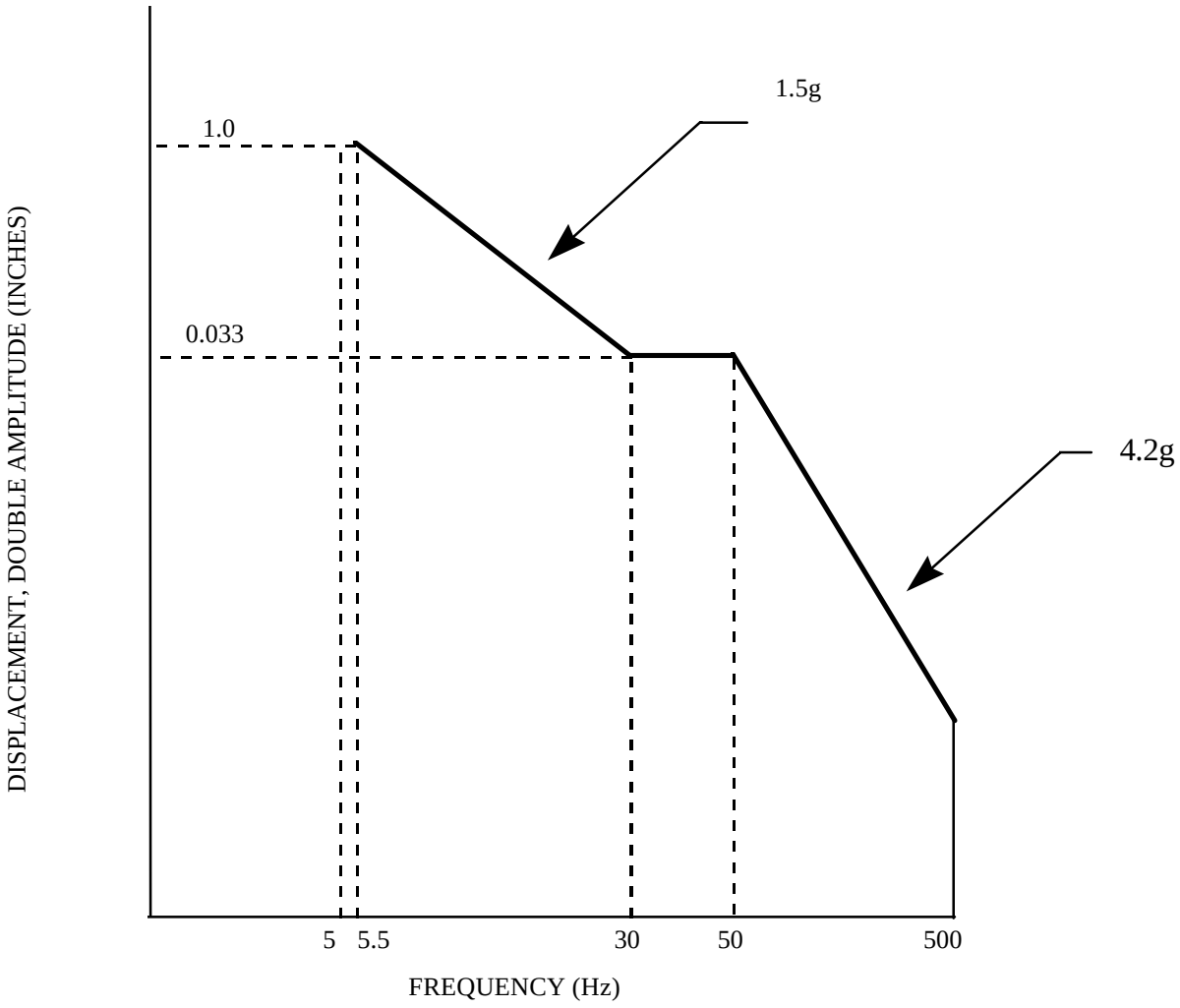


P - Peak value - 40 g
D = Nominal duration - 11 ms

NOTE: The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries, and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of $V_1 \pm 0.1V_1$ where V_1 is the velocity change associated with the ideal pulse which equals 0.5 PD. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

FIGURE 7. Terminal-peak sawtooth shock pulse configuration and tolerance limits.

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NOTES:

1. Acceleration levels: ± 9 g (peak).
2. The curve shall be extended to 2 Hz when test item resonances below 5 Hz are expected.

FIGURE 8. Vibration test curve.

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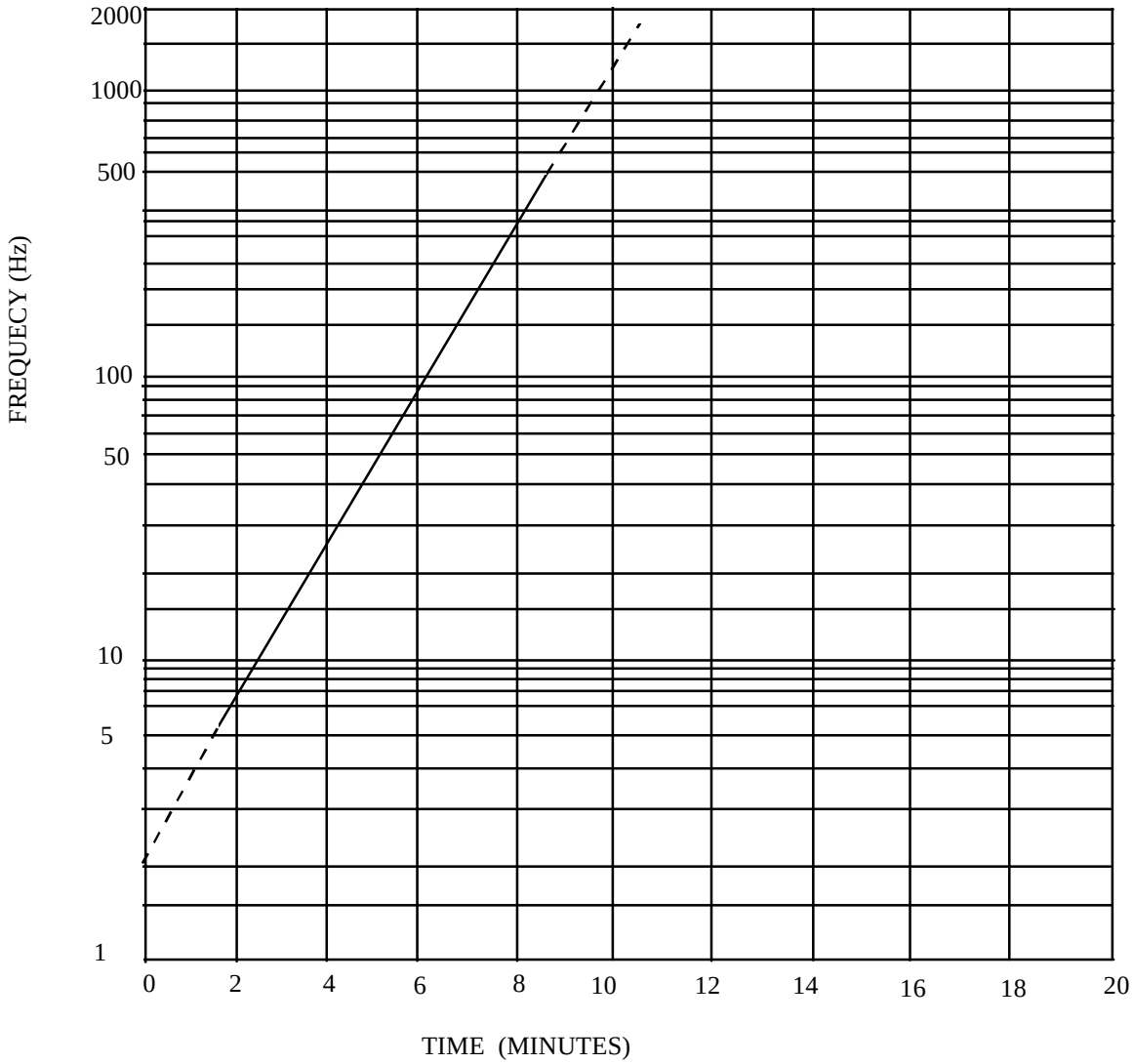


FIGURE 9. Logarithmic sweep.

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3. DOCUMENT TITLE
Air Cleaner, Engine

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

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a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
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